Indian Standard (Reaffirmed 1996)

SPECIFICATION FOR POWER TRANSFORMERS

PART I GENERAL

(First Revision)

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Indian Standard

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PART I GENERAL

(First Revision)

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Indian Standard

SPECIFICATION FOR POWER TRANSFORMERS

PART I GENERAL

(First Revision)

0. FOREWORD

0.1 This Indian Standard (Part I) (First Revision) was adopted by the Indian Standards Institution on 24 February 1977, after the draft finalized by the Transformers Sectional Committee had been approved by the Electrotechnical Division Council.

0.2 This standard was first published in 1962 and covered naturally-cooled oil-immersed transformers. By subsequent amendments forced-cooled transformers were included, use of synthetic liquids as cooling medium was permitted and requirements for aluminium windings were incorporated.

0.3 The revision has been undertaken with a view to bringing it in line with the revision of IEC Pub 76 - 1967 ' Power transformers '.

0.4 The salient features of this revision are as follows:

a) The kVA ratings have been extended to an unlimited range in the same series,
b) The requirements with regard to ability to withstand short circuit have been revised and a special test for assessing the dynamic ability of the transformer to withstand short circuit has been included,
c) Dry-type transformers have been covered,
d) Identification of transformers according to cooling methods has been lined up with the international practice, and
e) The phase marking *ABC* have been replaced by *UVW* following the international practice.

0.5 In this revision the requirements for power transformers are covered in four parts as follows:

Part I General
Part II Temperature-rise
Part III Insulation levels and dielectric tests
Part IV Terminal markings, tappings and connections

0.6 This standard (Part I) has been based on IEC Pub 76-1 (1976) Power transformers, Part I: General, issued by the International Electrotechnical Commission.

0.7 This part shall be read in conjunction with IS : 2026 (Part II) - 1977*, IS : 2026 (Part III) - 1977† and IS : 2026 (Part IV) - 1977‡.

0.8 For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS : 2 - 1960§. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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1. SCOPE

1.1 This standard covers power transformers (including autc transformers).

1.2 This standard does not cover the following small and special purpose transformers:

a) Single-phase transformers rated at less than 1 kVA and polyphase transformers rated at less than 5 kVA;

b) Outdoor type three-phase distribution transformers up to and including 100 kVA, 11 kV (covered by IS : 1180-1964‖);

c) Instrument transformers (covered by IS : 2705¶ and IS : 3156**);

d) Transformers for static convertors;

e) Starting transformers;

f) Testing transformers;

g) Traction transformers mounted on rolling stock;

h) Welding transformers [covered by IS : 1851-1966†† and IS : 4804 (Part I) - 1968‡‡];

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*Specification for power transformers: Part II Temperature-rise (*first revision*).

†Specification for power transformers: Part III Insulation levels and dielectric tests (*first revision*).

‡Specification for power transformers: Part IV Terminal markings, tappings and connections (*first revision*).

§Rules for rounding off numerical values (*revised*)

‖Specification for outdoor type three-phase distribution transformers up to and including 100 kVA 11 kV (*revised*).

¶Specification for current transformers.

**Specification for voltage transformers.

††Specification for single operator type arc welding transformers (*first revision*).

‡‡Specification for resistance welding equipment: Part I Single-phase transformers.

j) Mining transformers [covered by IS : 2772 (Part I) - 1964*];

k) Earthing transformers (covered by IS : 3151-1965†);

m) X-ray transformers;

n) Reactors (covered by IS : 5553‡); and

p) Furnace type transformers.

NOTE — Where Indian Standards do not exist for the transformers mentioned above, or for other special transformers, this standard may be applicable either as a whole or in part.

2. TERMINOLOGY

2.0 For the purpose of this standard, the definitions given in IS : 1885 (Part XXXVIII) - 1977§ shall apply.

3. SERVICE CONDITIONS

3.1 Reference Ambient Temperatures — The reference ambient temperatures assumed for the purpose of this specification are as follows:

a) Maximum ambient air temperature 50°C

b) Maximum daily average ambient air temperature 40°C

c) Maximum yearly weighted average ambient temperature 32°C

d) Minimum ambient air temperature — 5°C

e) *Water* — When the cooling medium is water, it is assumed that a temperature of 30°C will not be exceeded and that the average cooling water temperature will not exceed 25°C in any day.

3.1.1 Transformers complying with this specification are suitable for operation continuously, at their ratings provided the temperature of the cooling air or water does not exceed any of the reference ambient temperatures specified under **3.1**.

It is also recognized that operation of a transformer at its rated kVA provides normal life expectancy, if the hot spot temperature based on maximum yearly weighted average ambient temperature is 98°C.

3.1.2 For service conditions differing from standard, it is recommended that a transformer having a standard temperature rise be used and loaded in accordance with IS : 6600-1972‖.

*Specification for non-flameproof mining transformers for use below ground: Part I Oil-immersed type.

†Specification for earthing transformers.

‡Specification for reactors.

§Electrotechnical vocabulary: Part XXXVIII Transformers (*first revision*).

‖Guide for loading of oil-immersed transformers.

3.2 Altitude

3.2.1 Transformers conforming to this standard are suitable for operation at an altitude not exceeding 1 000 metres above mean sea level.

3.2.2 For altitudes exceeding 1 000 metres above mean sea level, an adjustment of the temperature-rise shall be necessary in accordance with IS : 2026 (Part II) - 1977*.

4. RATING

4.0 General — The manufacturer shall ascribe ratings to the transformer, which shall be marked on the rating plate (*see* **15**). These ratings shall be such that the transformer can deliver its rated current under steady loading conditions without exceeding the limits of temperature-rise specified in IS : 2026 (Part II) - 1977* assuming that the applied voltage is equal to the rated voltage and that the supply is at rated frequency.

4.1 Rated kVA — The rated kVA assigned shall take into account service conditions corresponding to those specified in 3 and shall be related to the product of rated voltage, rated current and the appropriate phase factor given in Table 1.

TABLE 1 PHASE FACTORS

Number of Phases	Phase Factor
(1)	(2)
1	1
3	1·73

NOTE 1 — The rated kVA assigned, corresponds to continuous duty; nevertheless oil-immersed transformers complying with this standard may be overloaded and guidance on overloads is given in IS : 6600-1972†.

Within the conditions defined in IS : 6600-1972†, occasional overloads‡ up to 1·5 times the rated value may be allowed on transformers with rated powers up to 100 MVA§. Under these conditions no limitations by bushings, tap-changers, or other auxiliary equipment shall apply. Regular daily overloads or emergency overloads in excess of this may be restricted by consideration of auxiliary equipment and in these cases reference shall be made to the manufacturer.

NOTE 2 — With rated voltage applied to one of the windings, the apparent power (kVA) that can really be delivered by (one of) the other winding(s) loaded with its rated current will deviate from its rated kVA by an amount depending on the corresponding voltage drop (or rise). This apparent power is equal to the product of the actual voltage on load of the latter winding, the rated current related to that winding and the appropriate phase factor (*see* Table I).

*Specification for power transformers: Part II Temperature rise (*first revision*).
†Guide for loading of oil immersed transformers.
‡Occasional overload implies:
a) operation at 1·5 times the rated current of the transformer for 3 percent of its life without making tap-changes, and
b) tap-changing on the basis of 3 percent of the operations being at currents of 1·5 times the rated current of the transformer.
§Special purpose transformers, for example, generator transformers, are not considered normally to require overload capacities.

4.2 kVA Ratings — kVA ratings for three-phase transformers are given in Table 2.

TABLE 2 kVA RATINGS FOR THREE-PHASE TRANSFORMERS

kVA	kVA	kVA
5	31·5	200
6·3	40	250
8	50	315
10	63	400
12·5	80	500
16	100	630
20	125	800
25	160	1 000 *et al*

NOTE — Underlined values are preferred ratings.

For single-phase transformers intended for use in three-phase banks, the kVA ratings are one-third of the values in Table 2; for single-phase transformers not intended for such use, they are the same as for three-phase transformers.

4.3 Operation at Other than Rated Voltage and Frequency

4.3.1 A transformer built in accordance with this specification may be operated at its rated kVA at any voltage within ± 10 percent of the rated voltage of that particular tap.

4.3.2 The transformer shall be capable of delivering rated current at a voltage equal to 105 percent of the rated voltage.

NOTE — The slight temperature-rise increase which would correspond to the 5 percent overvoltage, due to high no-load loss, is disregarded.

4.3.3 A transformer for two or more limits of voltage or frequency or both shall give its rated kVA under all the rated conditions of voltage or frequency or both; provided an increase in voltage is not accompanied by a decrease in frequency.

NOTE — Operation of a transformer at rated kVA at reduced voltage may give rise to excessive losses and temperature-rise.

4.4 Rated Frequency — The frequency for the purpose of this standard shall be 50 Hz with tolerance of ± 3 percent.

5. TEMPERATURE-RISE

5.1 The transformer shall conform to the requirements of temperature-rise specified in IS : 2026 (Part II)-1977*.

6. INSULATION LEVELS

6.1 The insulation levels shall be in accordance with IS : 2026 (Part III)-1977†.

7. TERMINAL MARKINGS, TAPPINGS AND CONNECTIONS

7.1 The terminal markings, tappings and connections shall be in accordance with IS : 2026 (Part IV)-1977‡.

8. REQUIREMENTS WITH REGARD TO ABILITY TO WITHSTAND SHORT CIRCUIT

8.0 General—Transformers shall be designed and constructed to withstand without damage the thermal and dynamic effects of external short circuit under the conditions specified in **8.1**.

NOTE — External short circuits are not restricted to three-phase short circuits, they include line-to-line, double line-to-earth and line-to-earth faults. The currents resulting from these conditions in the windings are designated as overcurrents in this standard.

8.1 Overcurrent Conditions

8.1.1 *Transformers with Two Separate Windings*

8.1.1.1 The following three categories for the rated kVA of three-phase transformers or three-phase banks shall be recognized:

Category 1 Up to 3 150 kVA

Category 2 3151 to 40 000 kVA

Category 3 Above 40 000 kVA

8.1.1.2 The symmetrical short-circuit current (rms value) (*see* **9.1.2**) shall be calculated using the short-circuit impedance of the transformer plus the system impedance for transformers of categories 2 and 3 and also for transformers of category 1 if the system impedance is greater than 5 percent of the short-circuit impedance of the transformer.

For transformers of category 1 the system impedance shall be neglected in the calculation if this impedance is equal to or less than 5 percent of the short-circuit impedance of the transformer.

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*Specification for power transformers: Part II Temperature-rise (*first revision*).

†Specification for power transformers: Part III Insulation levels and dielectric tests (*first revision*).

‡Specification for power transformers: Part IV Terminal markings, tappings and connections (*first revision*).

The peak value of the short-circuit current shall be calculated in accordance with **16.11.2**.

8.1.1.3 Typical values for the short-circuit impedance of transformers expressed as the impedance voltage at rated current (principal tapping) are given in Table 3. If lower values are required, the ability of the transformer to withstand a short circuit shall be subject to agreement between the manufacturer and the purchaser.

TABLE 3 TYPICAL VALUES OF IMPEDANCE VOLTAGE FOR TRANSFORMERS WITH TWO SEPARATE WINDINGS (AT RATED CURRENT, GIVEN AS A PERCENTAGE OF THE RATED VOLTAGE OF THE WINDING TO WHICH THE VOLTAGE IS APPLIED)

Rated Power	Impedance Voltage
(1)	(2)
kVA	Percent
Up to 630	4·0
631 to 1 250	5·0
1 251 ,, 3 150	6·25
3 151 ,, 6 300	7·15
6 301 ,, 12 500	8·35
12 501 ,, 25 000	10·0
25 001 ,, 200 000	12·5

NOTE 1 — Values for rated powers greater than 200 000 kVA shall be subject to agreement between the manufacturer and the purchaser.

NOTE 2 — In the case of single-phase transformers connected to form a three-phase bank, the value for rated power applies to the three-phase bank.

8.1.1.4 The short-circuit apparent power of the system at the transformer location shall be specified by the purchaser in his enquiry in order to obtain the value for the symmetrical short-circuit current to be used for the design and the tests. If the short-circuit level is not specified the value given in Table 4 shall be used.

8.1.2 *Transformers with More Than Two Windings and Auto-Transformers*

8.1.2.1 The overcurrents in the windings including stabilizing windings and auxiliary windings, shall be determined from the impedances of the transformer and the system(s). Account shall be taken of the effect of possible feedback from rotating machinery or from other transformers as well as of the different forms of system faults that can arise in service, such as line-to-earth faults and line-to-line faults associated with the relevant system and transformer earthing conditions. The characteristics of each system (at least the short-circuit level and the range of the ratio between the zero-sequence impedance and the positive sequence impedance) shall be specified by the purchaser in his enquiry.

TABLE 4 SHORT-CIRCUIT APPARENT POWER OF THE SYSTEM

(*Clauses* 8.1.1.4 *and* 9.1.2)

Highest System Voltage	Short-Circuit Apparent Power
(1)	(2)
kV	MVA
7·2, 12, 17·5 and 24	500
36	1 000
52 and 72·5	3 000
100 and 123	6 000
145	10 000
245	20 000
300	30 000
420	40 000

8.1.2.2 When the combined impedance of the transformer and the system(s) results in excessive overcurrent, the manufacturer shall advise the purchaser of the maximum overcurrent that the transformer can withstand. In this case, provision shall be made by the purchaser to limit the short-circuit current to the overcurrent indicated by the manufacturer.

8.1.2.3 Stabilizing windings of three-phase transformers shall be capable of withstanding the overcurrents resulting from different forms of system faults that can arise in service associated with relevant system earthing conditions.

8.1.2.4 It may not be economical to design auxiliary windings to withstand short circuits on their terminals. In such a case the effect of the overcurrents has to be limited by appropriate means, such as series reactors, or in some instances suitable fuses. Care has to be taken to guard against faults in the zone between the transformer and the protective apparatus.

8.1.2.5 In the case of single-phase transformers connected to form a three-phase bank the stabilizing windings shall be capable of withstanding a short circuit on their terminals, unless the purchaser specifies that special precautions shall be taken to avoid short circuits between phases.

8.1.3 *Booster-Transformers*

8.1.3.1 The impedances of booster-transformers may be very low and therefore the overcurrents in the windings are determined mainly by the characteristics of the system at the location of the transformer. These characteristics shall be specified by the purchaser in his enquiry.

8.1.3.2 When the combined impedance of the transformer and the system results in excessive overcurrent, the manufacturer shall advise the purchaser of the maximum overcurrent that the transformer can withstand. In this case, provision shall be made by the purchaser to limit the short-circuit current to the overcurrent indicated by the manufacturer.

8.1.4 *Transformers Directly Associated with Other Apparatus*

8.1.4.1 Where a transformer is directly associated with other apparatus, the impedance of which would limit the short-circuit current, the sum of the impedance of the transformer, the system and the directly associated apparatus shall, by agreement between manufacturer and purchaser be taken into account.

This applies, for example, to generator transformers if the connection between generator and transformer is constructed in such a way that the possibility of line-to-line or double line-to-earth faults in this region is negligible.

NOTE — The most severe short-circuit conditions may occur, in the case of a star-delta connected generator transformer with earthed neutral, when a line-to-earth fault occurs on the system connected to the star-connected winding.

8.1.5 *Special Transformers* — The ability of a transformer to withstand frequent overcurrents arising from the method of operation or the particular application (for example, furnace transformers, rectifier transformer and traction feeding transformers), shall be subject to special agreement between the manufacturer and the purchaser.

8.1.6 *Tap-Changing Equipment* — Where fitted, tap-changing equipment shall be capable of carrying the same overcurrents due to short circuits as the windings.

8.1.7 *Neutral Terminals* — The neutral terminal of windings with star or zigzag connection shall be designed for the highest overcurrent that can flow through this terminal.

9. DEMONSTRATION OF ABILITY TO WITHSTAND SHORT CIRCUIT

9.1 Thermal Ability to Withstand Short Circuit

9.1.1 *General* — The thermal ability to withstand short circuit shall be demonstrated by calculation.

9.1.2 *Value of the Symmetrical Short-Circuit Current* I *for Transformers with Two Windings* — The rms value of the symmetrical short-circuit current I is calculated for three-phase transformers as follows:

$$I = \frac{U}{(Z_t + Z_s)\sqrt{3}} \text{ kA}$$

where

Z_s is the short-circuit impedance of the system $\frac{U^2_s}{S}$ in ohms per phase:

U_s is the rated voltage of the system in kV and S is the short-circuit apparent power of the system in MVA;

U and Z_t are defined as follows:

a) for the principal tapping:

U is the rated voltage U_N of the winding under consideration, in kV;

Z_t is the short-circuit impedance of the transformer referred to the winding under consideration, and is calculated as follows:

$$Z_t = \frac{u_z \, U^2_N}{100 \, S_N}, \text{ in ohms per phase,}$$

where u_z is the impedance voltage at rated current and at the reference temperature, as a percentage, and S_N is the rated power of the transformer, in MVA; and

b) for tappings other than the principal tapping:

U is, unless otherwise specified, the tapping voltage of the tapping and the winding under consideration, in kV; and

Z_t is the short-circuit impedance of the tarnsformer referred to the winding and the tapping under consideration, in ohms per phase.

For transformers of category I the impedance of the system is neglected in the calculations if it is equal to or less than 5 percent of the short-circuit impedance of the transformer (*see also* **8.1.1.2**).

If the short-circuit power of the system is not specified by the purchaser in the enquiry, its value may be taken from Table 4.

9.1.3 *Duration of the Symmetrical Short-Circuit Current* I — The duration of the current I to be used for the calculation of the thermal ability to withstand short circuit shall be 2 seconds, unless otherwise specified by the purchaser.

NOTE — For auto-transformers and for transformers with a short-circuit current exceeding 25 times the rated current a short-circuit current duration below 2 seconds may be adopted after agreement between the manufacturer and the purchaser.

9.1.4 *Maximum Permissible Value of the Highest Average Temperature* θ_1 — On the basis of an initial winding temperature θ_0, derived from the sum of the maximum permissible ambient temperature and the relevant temperature rise at rated conditions measured by resistance (or, if this temperature-rise is not available, the temperature-rise for the relevant class of temperature of the winding) the highest average temperature θ_1 of the winding, after loading with a symmetrical short-circuit current I of a value and duration as described in **9.1.2** and **9.1.3** shall not exceed the value stated for θ_2 in Table 5 in any tapping position.

TABLE 5 MAXIMUM PERMISSIBLE VALUES OF AVERAGE TEMPERATURES θ_2 OF THE WINDING AFTER SHORT CIRCUIT

(*Clauses* 9.1.4, 9.1.5 *and* 9.1.6)

TRANSFORMER TYPE	CLASS OF TEMPERATURE	VALUE OF θ_2	
		Copper	Aluminium
(1)	(2)	(3)	(4)
		°C	°C
Oil-immersed	A	250	200
Dry	A	180	180
	E	250	200
	B	350	200
	F and H	350	—

9.1.5 *Calculation of the Temperature* θ_1 — The highest average temperature θ_1 attained by the winding after short circuit shall be calculated by the following formula:

$$\theta_1 = \theta_0 + a.J^2.t.\ 10^{-3}\text{°C},$$

where

θ_0 is the initial temperature in degrees Celsius;

J is the short-circuit current density in amperes per square millimetre;

t is the duration in seconds;

a is a function of $1/2\ (\theta_2 + \theta_0)$, in accordance with Table 6; and

θ_2 is the maximum permissible average winding temperature, in degrees Celsius, as specified in Table 5.

TABLE 6 VALUES OF FACTOR '*a*'

(*Clause* 9.1.5)

1/2 ($\theta_2 + \theta_0$)	*a**	
	Copper Windings	Aluminium Windings
(1)	(2)	(3)
°C		
140	7·41	16·5
160	7·80	17·4
180	8·20	18·3
200	8·59	19·1
220	8·99	—
240	9·38	—
260	9·78	—

*Function of 1/2 ($\theta_2 + \theta_0$).

9.1.6 *Value of the Symmetrical Short-Circuit Current* I *for Transformers with More Than Two Windings and Auto-Transformers* — The overcurrent shall be calculated in accordance with **8.1.2.** The highest average temperature of each winding shall be calculated in accordance with **9.1.3, 9.1.4** and **9.1.5** and shall not exceed the maximum permissible values given in Table 5.

9.2 The Dynamic Ability to Withstand Short Circuit

9.2.1 *General* — The dynamic ability to withstand short circuit shall be demonstrated by tests or by reference to tests on similar transformers.

Short-circuit tests are special tests carried out in accordance with **16.11.**

NOTE — Transformers of category 3 can normally not be tested in accordance with this standard. The conditions of tests on transformers with more than two windings and auto-transformers shall always be subject to agreement between the manufacturer and the purchaser.

10. MISCELLANEOUS REQUIREMENTS

10.1 Dimensions of Neutral Connection — The neutral conductor and terminal of transformers (for example, distribution transformers) intended to carry a load between phase and neutral shall be dimensioned for the appropriate load current and earth fault current.

The neutral conductor and terminal of transformers not intended to carry load between phase and neutral shall be dimensioned for earth fault current.

10.2 Load Rejection on Generator Transformers — Transformers intended to be connected directly to generators in such a way that they may be subjected to load rejection conditions shall be able to withstand 1·4 times the rated voltage for 5 seconds at the transformer terminals to which the generator is to be connected.

10.3 Insulation Liquids — Mineral oil, if used, shall comply with IS : 335-1972*.

10.4 Bushings — The bushings used on power transformers shall comply with IS : 2099-1973†.

10.4.1 The dimensions of bushings of the following voltage classes shall conform to Indian Standards mentioned against them:

Voltage Class	*Indian Standard*	
	For Porcelain Parts	For Metal Parts
Up to 1·1 kV	IS : 3347 (Part I/Sec 1)-1965‡	IS : 3347 (Part I/Sec 2)-1967§
3·6 kV	IS : 3347 (Part II/Sec 1)-1965‖	IS : 3347 (Part II/Sec 2)-1967¶
12 and 17·5 kV	IS : 3347 (Part III/Sec 1)-1972**	IS : 3347 (Part III/Sec 2)-1967††
24 kV	IS : 3347 (Part IV/Sec 1)-1973‡‡	IS : 3347 (Part IV/Sec 2)-1967§§
36 kV	IS : 3347 (Part V/Sec 1)-1973‖‖	IS : 3347 (Part V/Sec 2)-1967¶¶

*Specification for new insulating oils for transformers and switchgear (*second revision*).

†Specification for bushing for alternating voltages above 1 000 volts (*first revision*).

‡Dimensions for porcelain transformer bushings: Part I Up to 1·1 kV bushings, Section 1 Porcelain parts.

§Dimensions for porcelain transformer bushings: Part I Up to 1·1 kV bushings, Section 2 Metal parts.

‖Dimensions for porcelain transformer bushings: Part II 3·6 kV bushings, Section 1 Porcelain parts.

¶Dimensions for porcelain transformer bushings: Part II 3·6 kV bushings, Section 2 Metal parts.

**Dimensions for porcelain transformer bushings: Part III 12 and 17·5 kV bushings, Section 1 Porcelain parts (*first revision*).

††Dimensions for porcelain transformer bushings: Part III 12 and 17·5 kV bushings, Section 2 Metal parts.

‡‡Dimensions for porcelain transformer bushings: Part IV 24 kV bushings, Section 1 Porcelain parts (*first revision*).

§§Dimensions for porcelain transformer bushings: Part IV 24 kV bushings, Section 2 Metal parts.

‖‖Dimensions for porcelain transformer bushings: Part V 36 kV bushings, Section 1 Porcelain parts (*first revision*).

¶¶Dimensions for porcelain transformer bushings: Part V 36 kV bushings, Section 2 Metal parts.

11. TOLERANCES

11.1 Because of inevitable differences in basic materials and variations in manufacture, as well as measurement errors, the values obtained on test may differ from the calculated values and tolerances on guaranteed values are necessary.

Table 7 gives tolerances to be applied to certain rated quantities and to other quantities when they are the subject of manufacturers' guarantees referring to this standard. Where a tolerance in one direction is omitted there is no restriction on the value in that direction.

A transformer is considered as complying with this standard when the quantities subject to tolerances are not outside the tolerances given in Table 7.

12. INFORMATION REQUIRED WITH ENQUIRY AND ORDER

12.1 The technical information that the purchaser is required to supply with the enquiry and order is given in Appendix A.

13. TECHNICAL PARTICULARS TO BE FURNISHED BY THE MANUFACTURER

13.1 The manufacturer shall furnish technical particulars in accordance with Appendix B wherever required with the enquiry.

14. FITTINGS

14.1 Fittings as listed in Appendix C shall be provided.

15. RATING PLATES

15.1 Each transformer shall be provided with a rating plate of weatherproof material, fitted in a visible position, showing the appropriate items given in **15.2**. The entries on the rating plate shall be indelibly marked (for example, by etching, engraving or stamping).

15.2 Information to be Given in all Cases

a) Kind of transformer (for example, transformer, auto-transformer, booster transformer, etc);

b) Number of this standard, Ref IS : 2026;

c) Manufacturer's name;

d) Manufacturer's serial number;

e) Year of manufacture;

f) Number of phases;

g) Rated kVA (for multi-winding transformers the rated kVA of each winding shall be given. The loading combinations shall also be indicated unless the rated power of one of the windings is the sum of the rated powers of the other windings);

h) Rated frequency;

j) Rated voltages;

k) Rated currents;

m) Connection symbol;

n) Percent impedance voltage at rated current (measured value corrected to 75°C) and, if necessary, the reference power. The reference power should always be given for multi-winding transformers;

p) Type of cooling (if the transformer has several methods of cooling, outputs that differ from the rated power may be shown as percentages of the rated power, for example, ONAN/ONAF 70/100 percent);

q) Total mass; and

r) Mass and volume of insulating oil.

15.2.1 If the transformer has more than one rating, depending upon different connections of windings which have been specifically allowed for in the design, the additional ratings shall all be given on the rating plate.

15.3 Additional Information to be Given in Certain Cases

a) Temperature class of insulation (for dry-type transformers);

b) Temperature-rise (if other than that specified);

c) Connection diagram (in cases where the connection symbol will not give complete information regarding the internal connections). If the connections can be changed inside the transformer, the connection fitted at the works shall be indicated;

d) Insulation levels (applicable to windings rated for voltages of 3·6 kV and above and to the neutral end of windings with non-uniform insulation);

e) Transportation mass (for transformers exceeding 5 tonnes total mass);

f) Untanking mass (for transformers exceeding 5 tonnes total mass);

g) Insulating liquid, if not mineral oil;

NOTE — In the case of transformers immersed in synthetic insulating liquid containing polychlorinated biphenyls, for example Askarel, it is recommended that a distinctive plate be fitted with wording drawing attention to the need to take care because of environmental considerations.

h) Indication of the winding which is fitted with tappings;

j) Table giving the tapping voltage, the tapping current and the tapping power of each winding, for each tapping (*see* Note 1);

k) Values of short-circuit impedance on the extreme tappings and on the principal tapping and indication of the winding to which the impedance is related (*see* Note 1); and

m) Information on the ability of the transformer to operate at a voltage exceeding 105 percent of the tapping voltage or, for the principal tapping, 105 percent of the rated voltage [*see* IS : 2026 (Part IV)-1977* and **4.3**].

NOTE 1 — For separate winding transformers of rated power up to 3 150 kVA and tapping range up to ± 5 percent the table may be limited to the tapping voltages [it is then implied that tapping currents and tapping power vary as indicated in **3.2.3** of IS : 2026 (Part IV)-1977*] and the values of impedance may be limited to the value on the principal tapping.

NOTE 2 — More detailed information may be given on the rating plate or on a special plate, by agreement between the manufacturer and the purchaser. If necessary, a list of all characteristics should be given on a special plate, preferably in tabular form.

15.4 The power transformer may also be marked with the Standard Mark.

NOTE — The use of the Standard Mark is governed by the provisions of the Bureau of Indian Standards Act, 1986 and the Rules and Regulations made thereunder. The Standard Mark on products covered by an Indian Standard conveys the assurance that they have been produced to comply with the requirements of that standard under a well defined system of inspection, testing and quality control which is devised and supervised by BIS and operated by the producer. Standard marked products are also continuously checked by BIS for conformity to that standard as a further safeguard. Details of conditions under which a licence for the use of the Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

16. TESTS

16.1 General Requirements for Type, Routine and Special Tests — Tests shall be made at any ambient air temperature below 50°C and with cooling water (if required) at any temperature not exceeding 30°C.

Tests shall be made at the manufacturers' works, unless otherwise agreed between the manufacturer and the purchaser.

All external components and fittings that are likely to affect the performance of the transformer during the test shall be in place.

Tapped windings shall be connected on their principal tapping, unless the relevant test clause requires otherwise or unless the manufacturer and the purchaser agree otherwise.

*Specification for power transformers : Part IV Terminal markings, tappings and connections (*first revision*).

TABLE 7 TOLERANCES

(*Clause* 11.1)

Sl No.	Item	Tolerance
i)	a) Total losses	+ 10 percent of the total losses
	b) Component losses	+ 15 percent of each component loss, provided that the tolerance for total losses is not exceeded
ii)	Voltage ratio at no load on the principal tapping (rated voltage ratio) (*see* Note 2)	The lower of the following values: a) ± 0·5 percent of the declared ratio b) A percentage of the declared ratio equal to 1/10 of the actual percentage impedance voltage at rated current (*see* Note 3)
iii)	Impedance voltage at rated current (principal tapping):	
	a) If the principal tapping corresponds with the mean tapping position or with one of the two middle tapping positions:	
	1) two-winding transformers	± 10 percent of the declared impedance voltage for that tapping
	2) multi-winding transformers	± 10 percent of the declared impedance voltage for one specified pair of windings ± 15 percent of the declared impedance for a second specified pair of windings Tolerance to be agreed and stated for further pairs of windings
	b) For tappings other than the principal tapping	*See* IS : 2026 (Part IV)-1977*
iv)	Short-circuit impedance for any tapping	Not less than as indicated in [(iii)a] [*see also* IS : 2026 (Part IV)-1977*]
v)	No-load current	+ 30 percent of the declared no-load current

NOTE 1 — For Item (i) the loss tolerances of multi-winding transformers apply to every pair of windings unless the guarantee states that they apply to a given loading condition.

NOTE 2 — Tolerances on other tappings shall be subject to agreement between the manufacturer and the purchaser.

NOTE 3 — Alternative [(ii)b] does not apply in the case of auto-transformers and booster transformers, where the smallness of the impedance would result in an exceptionally small tolerance.

*Specification for power transformers: Part IV Terminal markings, tappings and connections (*first revision*).

The test basis for all characteristics, other than insulation, is the rated condition, unless the test clause states otherwise.

Where it is required, test results shall be corrected to a reference temperature of 75°C.

16.1.1 *Type Tests* — The following shall constitute the type tests:

a) Measurement of winding resistance (**16.2**);
b) Measurement of voltage ratio and check of voltage vector relationship (**16.3**);
c) Measurement of impedance voltage/short-circuit impedance (principal tapping) and load loss (**16.4**);
d) Measurement of no-load loss and current (**16.5**);
e) Measurement of insulation resistance (**16.6**);
f) Dielectric tests (**16.7**);
g) Temperature-rise (**16.8**); and
h) Tests on on-load tap-changers, where appropriate (**16.9**).

16.1.2 *Routine Tests* — The following shall constitute the routine tests:

a) Measurement of winding resistance (**16.2**);
b) Measurement of voltage ratio and check of voltage vector relationship (**16.3**);
c) Measurement of impedance voltage/short-circuit impedance (principal tapping) and load loss (**16.4**);
d) Measurement of no-load loss and current (**16.5**);
e) Measurement of insulation resistance (**16.6**);
f) Dielectric tests (**16.7**); and
g) Tests on on-load tap-changers, where appropriate (**16.9**).

16.1.3 *Special Tests* — The following tests shall be carried out by mutual agreement between the purchaser and the supplier:

a) Dielectric tests (**16.7**);
b) Measurement of zero-sequence impedance of three-phase transformers (**16.10**);
c) Short-circuit test (**16.11**);
d) Measurement of acoustic noise level (**16.12**);
e) Measurement of the harmonics of the no-load current (**16.13**); and
f) Measurement of the power taken by the fans and oil pumps (**16.14**).

16.1.3.1 If special tests other than those listed in **16.1.3** are required, the test method shall be subject to agreement between the manufacturer and the purchaser.

16.2 Measurement of Winding Resistance

16.2.1 *General* — The resistance of each winding, the terminals between which it is measured and the temperature of the windings shall be recorded. Direct current shall be used for the measurement.

In all resistance measurements care shall be taken that self-inductive effects are minimized.

During these cold-resistance measurements, the time for the measuring current to become steady should be noted and used for guidance when making hot-resistance measurements following a temperature-rise type test

16.2.2 *Dry-Type Transformers* — The temperature recorded shall be the average reading of several (at least three) thermometers placed on the winding surface.

Winding resistance and temperature shall be measured simultaneously, and the temperature of the winding, as measured by thermometer, should approximately equal the temperature of the surrounding medium.

16.2.3 *Oil-Immersed Type Transformers* — After the transformer has been under oil without excitation for at least 3 hours, the average oil temperature shall be determined and the temperature of the winding shall be deemed to be the same as the average oil temperature. The average oil temperature is taken as the mean of the top and bottom oil temperatures.

In measuring the cold resistance for pusposes of the temperature-rise test, special efforts shall be made to determine the average winding temperature accurately. Thus the difference in temperature between the top and bottom oil shall be small. To obtain this result more rapidly, the oil may be circulated by a pump.

16.3 Measurement of Voltage Ratio and Check of Voltage Vector Relationship — The voltage ratio shall be measured on each tapping. The polarity of single-phase transformers and vector symbol of three-phase transformers shall be checked.

16.4 Measurement of Impedance Voltage/Short-Circuit Impedance (Principal Tapping) and Load Loss — The impedance voltage/short-circuit impedance (principal tapping) and load loss shall be measured at rated frequency by applying an approximately sinusoidal supply to one winding, with the other winding short circuited, with the windings connected on the relevant tapping. The measurements may be

made at any current between 25 percent and 100 percent, but preferably not less than 50 percent, of the rated current (principal tapping) or tapping current. Each measurement shall be performed quickly and the intervals between them shall be long enough to ensure that temperature-rises do not cause significant errors. The difference in temperature between the top oil and the bottom oil shall be small enough to enable the average temperature to be determined with the required accuracy. If necessary the oil may be circulated by a pump.

The measured values of the load loss shall be corrected by multiplying them by the square of the ratio of rated current (principal tapping) or tapping current to test current. The value so derived shall be corrected to the reference temperature given in **16.1** taking the $I^2 R$ loss (R = dc resistance) as varying directly with resistance and all other losses as varying inversely with resistance. The resistance shall be determined as specified in **16.2**.

The measured value of the impedance voltage (principal tapping) shall be corrected by increasing it in the ratio of rated current to test current. The value of impedance voltage so derived shall be corrected to the reference temperature given in **16.1**.

The measured value of short-circuit impedance when given in ohms per phase, shall also be corrected to the reference temperature given in **16.1**.

On three-winding transformers the impedance voltage/short-circuit impedances (principal tapping) and the load losses shall be measured between windings taken in pairs as shown below:

a) Between winding 1 and winding 2
b) Between winding 2 and winding 3
c) Between winding 3 and winding 1

} the other winding being open-circuited.

For transformers with more than three windings, the windings shall be taken in pairs and the principle of the method specified for three-winding transformers shall be followed.

NOTE — The resistance of the test connections should be sufficiently low not to affect the measurement. If it is impracticable to employ connections in which the loss may be neglected in relation to the load loss of the transformer; allowance should be made for such losses.

16.5 Measurement of No-Load Loss and Current — The no-load loss and the no-load current shall be measured at rated frequency at a voltage equal to rated voltage if the test is performed on the principal tapping or equal to the appropriate tapping voltage if the test is performed on another tapping. Other winding(s) shall be left open-circuited and any windings which may be connected in open-delta shall have the delta closed.

For all transformers, the voltage shall be measured with a voltmeter responsive to the mean value of voltage but scaled to read the rms value of a sinusoidal wave having the same mean value. The voltage U' indicated by this voltmeter shall be taken as the required value of line-to-line voltage and the no-load loss P_m shall be measured at this voltage.

At the same time, a voltmeter responsive to the rms value of voltage shall be connected in parallel with the mean-value voltmeter and its indicated voltage U shall be recorded.

If the voltages U' and U are not the same, the measured value of no-load loss is corrected according to the formula:

$$P = \frac{P_m}{P_1 + kP_2}$$

where

P = corrected value,

P_1 = ratio of hysteresis losses to total iron losses,

$k = \left(\frac{U}{U'}\right)^2$, and

P_2 = ratio of eddy current losses to total iron loss.

NOTE — For flux densities normally used at 50 or 60 Hz the following values should be taken:

	P_1	P_2
Oriented steel	0·5	0·5
Non-oriented steel	0·7	0·3

The no-load current of all the phases shall be measured by rms ammeters, and the mean of their readings shall be taken as the no-load current.

16.6 Measurement of Insulation Resistance — The oil/air temperatures shall be measured and recorded immediately prior to the test. The insulation resistance of each winding, in turn, to all the other windings, core and frame or tank connected together, and to earth shall be measured and recorded.

16.7 Dielectric Tests — The transformer shall pass the appropriate dielectric tests specified in IS : 2026 (Part III) - 1977*. The dielectric tests may be type tests, routine tests or special tests in accordance with IS : 2026 (Part III) - 1977*.

16.8 Temperature-Rise — The transformer shall pass the temperature-rise test specified in IS : 2026 (Part II) - 1977†.

*Specification for power transformers: Part III Insulation levels and dielectric tests (*first revision*).

†Specification for power transformers: Part II Temperature-rise (*first revision*).

16.9 Tests on On-Load Tap-Changers

16.9.1 *Operation Test* — After the tap-changer is fully assembled on the transformer, the transformer manufacturer shall perform the following tests [with the exception of (b)] at 100 percent of the rated auxiliary supply voltage. These tests shall be performed without failure.

a) Eight complete operating cycles with the transformer not energized;

b) One complete operating cycle with the transformer not energized, with 85 percent of the rated auxiliary supply voltage;

c) One complete operating cycle with the transformer energized at rated voltage and frequency at no load; and

d) Ten tap-change operations with ± 2 steps on either side of the principal tapping with as far as possible the rated current of the transformer with one winding short-circuited.

16.9.2 *Auxiliary Circuits Insulation Test* — After the tap-changer is assembled on the transformer, a power frequency test shall be applied to the auxiliary circuits as specified in IS: 2026 (Part III) - 1977*.

16.10 Measurement of Zero-Sequence Impedance(s) of Three Phase Transformers — The zero-sequence impedance is measured at rated frequency between the line terminals of a star-connected or zigzag-connected winding connected together and its neutral terminal. It is expressed in ohms per phase and is given by 3 U/I where U is the test voltage and I is the test current.

The phase test current $I/3$ shall be stated.

It shall be ensured that the current in the neutral connection is compatible with its current carrying capability.

In the case of a transformer with an additional delta-connected winding, the value of the test current shall be such that the current in the delta-connected winding is not excessive, taking into account the time of application.

If balancing ampere-turns are missing in the zero-sequence system, for example, in a star-star-connected transformer without delta winding, the applied voltage shall not exceed the line-to-neutral voltage at normal operation.

The current in the neutral and the time of application however may be limited to avoid excessive temperatures of metallic constructional parts.

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*Specification for power transformers: Part III Insulation levels and dielectric tests (*first revision*).

In the case of transformers with more than one star-connected winding and neutral terminals, the zero-sequence impedance [*see* **2.7.5** of IS : 1885 (Part XXXVIII) - 1977*] is dependent upon the connections and the tests to be made shall be subject to agreement between the manufacturer and the purchaser.

Measurements on windings with tappings shall be made on the principal tapping. Measurements on other tappings may be made if agreed between the manufacturer and the purchaser.

Auto-transformers with a neutral terminal intended to be permanently connected to earth shall be treated as normal transformers with two star-connected windings. Thereby the series winding and the common winding together form one measuring circuit, and the common winding alone the other. The measurements shall be carried out with a current not exceeding the difference between the rated currents on the low-voltage side and the high-voltage side.

NOTE 1 — In conditions where the balancing ampere-turns are missing the relation between voltage and current is generally not linear.

NOTE 2 — The zero-sequence impedance is dependent upon the physical disposition of the windings and magnetic parts and measurements on different windings may not therefore agree.

16.11 Short-Circuit Test

16.11.1 *Condition of the Transformers Before the Short-Circuit Tests*

16.11.1.1 Unless otherwise agreed, the tests shall be carried out new transformer ready for service. The mounting of accessories having no influence on the behaviour during short circuit (for example, detachable cooling equipment) is not required.

16.11.1.2 Prior to the short-circuit tests the transformer shall be subjected to the routine tests specified in **16.1.2**.

If the windings are provided with tappings the reactance and, if required, the resistance shall also be measured for the tapping positions at which short-circuit tests shall be carried out.

All the reactance measurements shall be to a reproducibility of at least ± 0·2 percent.

A report containing the results of the routine tests shall be available at the beginning of the short-circuit tests.

16.11.1.3 At the beginning of the short-circuit test the average temperature of the winding shall be between 0°C and 50°C.

*Electrotechnical vocabulary : Part XXXVIII Transformers (*first revision*).

16.11.2 *Peak Value $\hat{\imath}$ for Two-Winding Transformers* — The amplitude $\hat{\imath}$ of the first peak of the asymmetrical test current is calculated as follows:

$$\hat{\imath} = Ik\sqrt{2}$$

where the symmetrical short-circuit current I is determind in accordance with **8.1.1.2** and **9.1.2**.

The factor $k\sqrt{2}$ depends on the ratio X/R

where

X is the sum of the reactances of the transformer and the system ($X_t + X_s$), in ohms; and

R is the sum of the resistances of the transformer and the system ($R_t + R_s$), in ohms.

Unless otherwise specified the factor $k\sqrt{2}$ is limited to $1{\cdot}8\sqrt{2} = 2{\cdot}55$.

The following values of factor $k\sqrt{2}$ shall be used for different values of X/R:

X/R	1	1·5	2	3	4	5	6	8	10	≥14
$k\sqrt{2}$	1·51	1·64	1·76	1·95	2·09	2·19	2·27	2·38	2·46	2·55

NOTE — For other values of X/R between 1 and 14, the factor $k\sqrt{2}$ may be determined by linear interpolation.

For transformers of category 1 and $Z_s \leqslant 0{\cdot}05\ Z_t$ (*see* **8.1.1.2** and **9.1.2**) X and R are related to the transformer only (X_t and R_t). For $Z_s > 0{\cdot}05\ Z_t$, on the other hand, X and R are related to the transformer and the system ($X_t + X_s$ and $R_t + R_s$).

NOTE — When $Z_s \leqslant 0{\cdot}05\ Z_t$, instead of X_t and R_t (in ohms) u_x and u_r may be used for the principal tapping, where

u_x is the reactive component of u_z, in percent;

u_r is the resistance component, at reference temperature, of u_z, in percent; and

u_z is the impedance voltage of the transformer, at reference temperature, in percent.

16.11.3 *Value and Duration of the Short-Circuit Test Current for Two-Winding Transformers* — The asymmetrical current having a first peak of amplitude $\hat{\imath}$ (**16.11.2**) will change (if the duration of the test current is sufficiently long) into the symmetrical current I (**9.1.2**).

The peak value of the current obtained in testing shall not deviate by more than 5 percent and the symmetrical current by more than 10 percent from the specified value. The duration of the current for shortcircuit tests is specified in **16.11.4.4**.

16.11.4 *Short-Circuit Testing Procedure for Transformers with Two Windings* —

16.11.4.1 In order to obtain a test current according to **16.11.3** the no-load voltage of the source may be higher than the rated voltage of the winding supplied. The short-circuiting of the winding may either follow or precede (pre-set short-circuit) the application of the voltage to the other winding of the transformer.

In the first case the voltage shall not exceed 1·15 times the rated voltage of the winding.

If the pre-set short circuit is used for transformers with single concentric windings, the supply should be connected to the winding farther from the core, the winding nearer to the core being short-circuited in order to avoid saturation of the magnetic core, which could result in an excessive flow of magnetizing current being superimposed on the short-circuit current during the first few cycles.

For transformers with sandwich windings or transformers with double concentric windings the pre-set short-circuit method shall only be used after agreement between the manufacturer and the purchaser.

16.11.4.2 To obtain the initial peak value of the current (**16.11.2**) in the phase winding under test the moment of switching-on shall be adjustable by a synchronous switch.

In order to check the value of the test currents $\hat{\imath}$ and I these currents shall always be recorded by an oscillogragh. In order to obtain the maximum asymmetry of the current in one of the phase windings the switching-on shall occur at the moment the voltage of this winding passes through zero.

NOTE 1 — For star-connected windings the maximum asymmetry is obtained by switching-on when the phase voltage passes through zero. The factor k of the peak value $\hat{\imath}$ can be determined from the oscillograms of the line currents. For three-phase tests on delta-connected windings this condition is obtained by switching-on when the line-to-line voltage passes through zero. One of the methods of determining the factor k is by switching-on during the preliminary adjustment tests at a maximum of the line-to-line voltage. In this case the factor k is found from the oscillograms of the line currents.

Another method for determining the phase currents in a delta connected winding is by suitably interconnecting the secondary windings of the current transformers measuring the line currents. The oscillograph can be made to record the phase currents.

NOTE 2 — For transformers with star-zigzag connection belonging to category 1 and with constant flux voltage variation, having a value for $u_x/u_r \leqslant 3$ (*see* **16.11.2**), the three phases are switched-on simultaneously without the use of a synchronous switch. For other transformers with star-zigzag connection the method of switching on is subject to agreement between the manufacturer and the purchaser.

16.11.4.3 For three-phase transformers, a three-phase supply should be used, as long as the requirements given in **16.11.3** can be met. If this is not the case a single-phase supply, as described below, may be used. For delta-connected windings the single-phase supply is provided between two corners of the delta and the voltage during the test has to be the same as the voltage between phases during a three-phase test. For star-connected windings the single-phase voltage is supplied between one phase terminal and the other two phase terminals connected together. The single-phase voltage during the test has to be equal to $\sqrt{3/2}$ times the voltage between phases during a three-phase test.

NOTE 1 — The use of the tests with single-phase supply applies mainly to transformers of category 2 or 3 and is seldom of interest for category 1 transformers.

NOTE 2 — For star-connected windings with non-uniform insulation it is necessary to check whether or not the insulation of the neutral is sufficient for single-phase testing.

NOTE 3 — If for star-connected windings the power supply is insufficient for the single-phase testing described above and the neutral is available, the manufacturer and the purchaser may agree upon the use of single-phase tests between a line terminal and the neutral.

16.11.4.4 In the absence of any particular specification the number of tests on three-phase and single-phase transformers is determined as follows, not including preliminary adjustment tests carried out at less than 70 percent of the specified current to check the proper functioning of the test set-up with regard to the moment of switching-on, the current setting, the damping and the duration.

For category 1 single-phase transformers the number of tests shall be three, the duration of each test being 0·5 second with a tolerance of ± 10 percent. Unless otherwise specified each of the three tests on a single-phase transformer with tappings is made in a different position of the tap-changer, that is, one test in the position corresponding to the highest voltage ratio, one test on the principal tapping and one test in the position corresponding to the lowest voltage ratio.

For category 1 three-phase transformers the total number of tests shall be nine, that is three tests on each limb, the duration of each test being 0·5 second with a tolerance of ± 10 percent. Unless otherwise specified the tests on each limb of a transformer with tappings are made in different positions of the tap-changer, that is, three tests in the position corresponding to the highest voltage ratio on one of the outer limbs, three tests on the principal tapping on the middle limb and three tests in the position corresponding to the lowest voltage ratio on the other outer limb.

For transformers of categories 2 and 3, an agreement between the manufacturer and the purchaser is always needed with regard to the number of tests, their duration and the position of the tap-changer.

16.11.5 *Detection of Faults*

16.11.5.1 Before the short-circuit testing, measurements and tests shall be carried out according to **16.11.1** and the gas relay (if any) inspected. These measurements and tests serve as references for the detection of faults.

16.11.5.2 During each test (iucluding preliminary tests) oscillographic recording shall be made of the following:

a) The applied voltage (between line terminals), and

b) The currents (*see* Notes 1 and 2 under **16.11.4.2**).

Furthermore, the transformer under test shall be visually inspected.

NOTE — Additional means of detection may be used, such as oscillographic recording of the radial stray flux by means of supplementary coils, information obtained from the noise and, particularly, recording of the current between the tank (insulated) and earth.

16.11.5.3 After each test the oscillograms made during the test are inspected, and also the gas relay. It is necessary to measure the short-circuit reactance after each test.

NOTE 1 — Additional means of detection may be used, such as resistance measurements, oscillograms of impulse voltages for comparison with oscillograms obtained in the original state (recurrent surge oscillograph method) and a no-load measurement (for detection of a short circuit between turns).

NOTE 2 — Any differences between the results of the measurements made before and after the test may be a criterion for determining possible defects. It is of particular importance to observe during successive tests possible changes in the reactance measured after each test, which may be progressive or tending to a stable value.

16.11.5.4 After completion of the test the transformer and the gas relay, if any, shall be inspected. The results of the short-circuit reactance measurements and the oscillograms made during the different stages of the tests shall be compared.

NOTE — After agreement between the manufacturer and the purchaser all or a part of the routine tests (power frequency dielectric tests at 75 percent of the original value) may be performed at this stage.

Then the transformer is untanked for inspection of the core and windings.

If the results of the tests, measurements and inspection do not reveal any defect, the transformer shall be re-tanked, and the routine tests performed. The power frequency dielectric tests shall be made with 75 percent of the original test value unless a higher value has been agreed for the repeat tests. The inpulse tests are only repeated after agreement between the manufacturer and the purchaser.

If there is doubt or disagreement with regard to the result of tests or the inspection of the untanked transformer, a detailed examination, including complete dismantling of the unit is necessary. For category 1 transformers the complete dismantling is necessary when the reactance measured after the tests differs more than 2 percent from that measured in the original state. For the interpretation of this difference (if any) with regard to transformers of categories 2 and 3 agreement between the manufacturer and the purchaser is needed.

16.11.6 *Evaluation of the Results of Short-Circuit Tests* — The transformer shall be deemed to have withstood the short-circuit tests if the measurements and inspections (untanked or dismantled) show no apparent defects (displacements, deformations of windings, connections or supporting structures, or traces of discharge).

16.12 Measurement of Acoustic Noise Level — The method of test and criteria for conformity shall be agreed between the manufacturer and the purchaser.

16.13 Measurement of the Harmonics of the No-Load Current — The harmonics of the no-load current in all the phases are measured by means of harmonic analyser and the magnitude of the harmonics is expressed as a percentage of the fundamental component.

16.14 Measurement of Power Taken by the Fans and Oil-Pumps — The method of test shall be agreed between the manufacturer and the purchaser [*see also* **4.7** of IS: 2026 (Part II)-1977*].

APPENDIX A

(*Clause* 12.1)

INFORMATION REQUIRED WITH ENQUIRY AND ORDER

A-1. RATING AND GENERAL DATA

A-1.1 Two-Winding Transformers

A-1.1.1 *Normal* — The following information should be given in all cases:

a) Particulars of the specification to be complied with;

b) Type of transformer, for example, separate winding transformer; auto-transformer or booster transformer;

c) Single or polyphase unit;

*Specification for power transformers: Part II Temperature-rise (*first revision*).

d) Number of phases in system;

e) Frequency;

f) Dry-type or oil-immersed type. If oil-immersed, whether mineral oil or synthetic insulating liquid;

g) Indoor or outdoor type;

h) Type of cooling;

j) Rated power (in kVA) and, for tapping ranges exceeding ± 5 percent required power on extreme tappings;

If the transformer is to have different values of rated power for different types of cooling, these should be stated.

k) Rated voltages (for each winding);

m) State if tappings are required and if on-load or off-circuit tap-changers, or links are required. Give the information required as detailed in IS : 2026 (Part IV)-1977*;

n) Highest voltage for equipment (for each winding) [*see also* Appendix C of IS : 2026 (Part III)-1977†];

p) Method of system earthing (for each winding);

q) Insulation level (for each winding), power frequency test level/ impulse level [*see* IS : 2026 (Part III)-1977†];

r) Connection symbol;

s) Neutral terminals, if required (for each winding) and their insulation level to earth;

t) Special requirements of installation, assembly, transport and handling;

u) Details of auxiliary supply voltage (for fans, tap-changer, alarms, etc);

w) Fittings required and an indication of the side from which meters, rating plates, oil-level indicator, etc, may be readable.

A-1.1.2 *Special* — The following additional information may be required to be given:

a) If a lightning impulse voltage test is required, whether or not the test is to include chopped waves [*see* IS : 2026 (Part III)-1977†];

b) Whether a stabilizing winding is required and, if so, the method of earthing;

c) Impedance voltage at rated current, if specific value is required;

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*Specification for power transformers: Part IV Terminal markings, tappings and connections (*first revision*).

†Specification for power transformers: Part III Insulation levels and dielectric tests (*first revision*)

d) Whether transformer is to be connected to a generator directly or through switchgear and whether it mav be subjected to load rejection conditions;

e) Altitude above mean sea-level, if in excess of 1 000 m;

f) Special cooling conditions, for example, temperature of cooling medium if above or below values given in **3**, or restrictions to circulation of cooling air;

g) Whether unbalanced loading is anticipated, and if so details;

h) Whether transformers will be subjected to frequent overcurrents, for example, furnace transformers and traction feeding transformers (*see* **8**);

j) Details of intended regular cyclic overloading other than covered by **4.1** (to enable the ratings of the transformer auxiliary equipment to be established);

k) Any other exceptional service conditions;

m) Connections required, ex-works, in the case of the transformer having alternative winding connections;

n) Val e of load and no-load losses, if specific values are required;

p) Sh t-circuit characteristics of the system (*see* **8** and **9**);

q) Whether noise level measurement is to be carried out;

r) Vacuum withstand of the transformer tank, if a specific value is required;

s) Type of tap-changer controls required;

t) Water analysis in case of OFWF cooled transformers; and

u) Any other appropriate information, including reference to any special tests not referred to above which may be required.

A-1.2 Multi-Winding Transformers — Generally as in **A-1.1**, but information in respect of the additional windings is required, and also:

a) required loading combinations, stating, when necessary, the active and reactive outputs separately, especially in the case of multi-winding auto-transformer; and

b) any impedance voltages that are specified for particular pairs of windings.

A-2. PARALLEL OPERATION

A-2.1 If parallel operation with existing transformers is required, this should be stated and the following information given:

a) Rated kVA of existing transformers,

b) Rated voltage ratio,

c) Voltage ratios corresponding to tappings other than the principal tapping,

d) Load loss at rated current and rated voltage on the principal tapping corrected to the appropriate reference temperature,

e) Impedance voltage at rated current (on the principal tapping),

f) Short-circuit impedances at least on the extreme tappings if the tapping range of the tapped winding exceeds ± 5 percent, and

g) Diagram of connections and connection symbol.

APPENDIX B

(*Clause* 13.1)

SCHEDULE OF TECHNICAL PARTICULARS TO BE FURNISHED BY THE MANUFACTURER

B-1. STANDARD FORM OF TECHNICAL PARTICULARS

1. Name of the manufacturer
2. Service
3. kVA rating:
   a) HV winding kVA
   b) IV winding kVA
   c) LV winding kVA
4. Rated voltage:
   a) HV winding kV
   b) IV winding kV
   c) LV winding kV
5. Rated frequency Hz
6. Number of phases
7. Connections:
   a) HV winding
   b) IV winding
   c) LV winding

8. Connection symbol [*see* IS : 2026 (Part IV)-1977*]:
   a) HV-IV
   b) HV-LV
9. Tappings:
   a) Range
   b) Number of steps
   c) For high voltage variation/For intermediate voltage variation/For low voltage variation
10. Reference ambient temperatures:
   a) Maximum ambient air temperature °C
   b) Maximum daily average ambient air temperature °C
   c) Maximum yearly weighted average ambient temperature °C
   d) Minimum ambient air temperature °C
   e) Maximum cooling water temperature °C
11. Type of cooling [*see* IS : 2026 (Part II)-1977†]
12. Temperature-rise [*see* IS : 2026 (Part II)-1977†]:
   a) Top oil °C
   b) Windings °C
13. Total loss at rated voltage at principal tapping and rated frequency kW
14. Component losses:
   a) No load loss at rated voltage on principal tapping and at rated frequency kW
   b) Load loss at rated current at principal tapping at 75°C kW
15. Impedance voltage at rated current for the principal tapping:
   a) HV - IV percent
   b) HV - LV percent
   c) IV - LV percent

NOTE — Item 11 to 15 are guaranteed particulars and are subject to tolerances as specified in Table 7.

*Specification for power transformers: Part IV Terminal markings, tappings and connections (*first revision*).

†Specification for power transformers: Part II Temperature rise (*first revision*).

16. Reactance at rated current and rated frequency:
    a) HV - IV — percent
    b) HV - LV — percent
    c) IV - LV — percent
17. No load current at rated voltage and rated frequency — percent
18. Input to cooling plant — kW
19. Insulation level [*see* IS : 2026 (Part III)-1977*]:
    a) Separate source power-frequency voltage withstand:
        i) HV winding — kV rms
        ii) IV winding — kV rms
        iii) LV winding — kV rms
    b) Induced overvoltage withstand:
        i) HV winding — kV rms
        ii) IV winding — kV rms
        iii) LV winding — kV rms
    c) Full wave lightning impulse withstand voltage:
        i) HV winding — kV peak
        ii) IV winding — kV peak
        iii) LV winding — kV peak
    d) Switching impulse withstand voltage:
        i) HV winding — kV peak
        ii) IV winding — kV peak
        iii) LV winding — kV peak
20. Stabilizing/Tertiary winding:
    a) Rated voltage — kV
    b) Normal rating — kVA
    c) Normal rating (expressed as percent of main winding rating in the case of stabilizing winding) — percent
    d) Delta closed inside/outside (In the case of stabilizing winding)

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*Specification for power transformers: Part III Insulation levels and dielectric tests (*first revision*).

21. Efficiencies at 75°C at unity power factor:
    a) At full load percent
    b) At 3/4 full load percent
    c) At 1/2 full load percent
22. Regulation at full load at 75°C:
    a) At unity power factor percent
    b) At 0·8 power factor lagging percent
23. Equipment for ONAN or ONAF/ONAN Cooling:
    a) State:
        i) Radiators on main tank
        ii) Separate cooler bank
    b) State ONAN rating in case of mixed cooling
24. Equipment for OFAF/ODAF cooling:
    a) State:
        i) Radiator bank
        ii) Separate enclosed type coolers
    b) State OFAN/ODAN, ONAF, ONAN rating (kVA) whichever is/are applicable in case of mixed cooling:
        i) OFAN/ODAN
        ii) ONAF
        iii) ONAN
25. Number of coolers or cooler banks per transformer.
26. Rating of each cooler or cooler bank
27. In case of OFF circuit voltage variation, state (a) or (b):
    a) Off circuit tap switch
    b) Off circuit links
28. Details of on-load tap-changer:
    a) Make
    b) Type
    c) Rating:
        i) Rated voltage
        ii) Rated current
        iii) Step voltage
        iv) Number of steps

d) Control
e) Auxiliary supply details
f) Voltage control
g) Parallel operation
h) Protective devices
j) Approximate overall weight kg
k) Approximate overall dimensions mm
m) Approximate overall quantity of oil litre

NOTE — The information against (j), (k) and (m) is required in case of separately mounted and handled tap-changers.

29. Terminal arrangement:

a) High voltage
b) Intermediate voltage
c) Low voltage
d) Neutral

30. Approximate masses:

a) Core and windings kg
b) Tank, fittings and accessories kg
c) Oil kg
d) Total mass kg

31. Approximate quantity of oil required for first filling litres

32. Approximate overall demensions:

a) Length mm
b) Breadth mm
c) Height mm

33. Despatch details:

a) Approximate mass of heaviest package kg
b) Approximate dimensions of largest package:

i) Length mm
ii) Breadth mm
iii) Height mm

34. Untanking height mm

35. Reference standards

APPENDIX C

(*Clause* 14.1)

FITTINGS

C-1. The fittings as given below shall be provided:

Sl No.	*Item*	*Transformers to Which Fitted*
1.	Inspection cover	Above 1 600 kVA
2.	Rating plate	All
3.	Terminal-marking plate	All
4.	Two earthing terminals	All
5.	Lifting lugs	All
6.	Drain valve with plug or cover plate	25 kVA and above
7.	Dehydrating breather	25 kVA and above for rated voltage 11 kV and below, and all ratings above 11 kV
8.	Oil-level indicator with minimum	
9.	Marking	All
10.	Thermometer pocket	All
11.	Oil-filling hole with cover	All
12.	Conservator	50 kVA and above for rated voltage 11 kV and below, and all ratings above 11 kV
13.	Air release device	All transformers fitted with conservators
14.	Jacking lugs	Above 1 600 kVA
15.	Filter valve	Above 1 600 kVA

NOTE — When filter valves are required, the drain valve should be used as one of these.

C-2. The additional fittings as given below may be provided, if ordered:

Sl No.	*Item*	*Remarks*
1.	Filter valves	1 600 kVA and below
2.	Rollers	Details of rollers to be stated with enquiry or order
3.	Thermometer	Type to be stated with enquiry or order
4.	Additional thermometer pockets	To be stated with enquiry or order
5.	Winding temperature indicator	Details of requirments to be stated with enquiry or order
6.	Explosion vent	Details to be stated
7.	Gas- and oil-actuated relay	Type to be stated
8.	Skids	
9.	Inspection cover	1 600 kVA and below

NOTE — The fittings listed in **C-1** and **C-2** shall conform to relevant Indian Standards wherever they exist.

BUREAU OF INDIAN STANDARDS

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*Eastern : 1/14 CIT Scheme VII M, V.I.P. Road, Maniktola, CALCUTTA 700054	37 86 62
Northern : SCO 335-336, Sector 34-A, CHANDIGARH 160022	60 38 43
Southern : C.I.T. Campus, IV Cross Road, MADRAS 600113	235 23 15
†Western : Manakalaya, E9 MIDC, Marol, Andheri (East), BOMBAY 400093	832 92 95
Branch Offices :	
'Pushpak', Nurmohamed Shaikh Marg, Khanpur, AHMADABAD 380001	30 13 48
‡Peenya Industrial Area, 1st Stage, Bangalore-Tumkur Road, BANGALORE 560058	839 49 55
Gangotri Complex, 5th Floor, Bhadbhada Road, T.T. Nagar, BHOPAL 462003	55 40 21
Plot No. 21 Satyanagar, BHUBANESHWAR 751007	40 36 27
Kalaikathir Building, 6/48 Avanashi Road, COIMBATORE 641037	21 01 41
Plot No. 43, Sector 16 A, Mathura Road, FARIDABAD 121001	8-28 88 01
Savitri Complex, 116 G.T. Road, GHAZIABAD 201001	8-71 19 96
53/5 Ward No. 29, R.G. Barua Road, 5th By-lane, GUWAHATI 781003	54 11 37
5-8-56C L.N. Gupta Marg, Nampally Station Road, HYDERABAD 500001	20 10 83
R 14, Yudhister Marg, C Scheme, JAIPUR 302005	38 13 74
117/418 B Sarvodaya Nagar, KANPUR 208005	21 68 76
Seth Bhawan, 2nd Floor, Behind Leela Cinema, Naval Kishore Road, LUCKNOW 226001	23 89 23
Patliputra Industrial Estate, PATNA 800013	26 23 05
T.C. No. 14/1421, University P.O., Palayam, THIRUVANANTHAPURAM 695034	6 21 17
Inspection Offices (With Sale Point) :	
Pushpanjali, 1st floor, 205-A, West High Court Road, Shankar Nagar Square, NAGPUR 440010	52 51 71
Institution of Engineers (India) Building 1332 Shivaji Nagar, PUNE 411005	32 36 35

* Sales Office is at 5 Chowringhee Approach, P.O. Princep Street, CALCUTTA 700072	27 10 85
†Sales Office is at Novelty Chambers, Grant Road, BOMBAY 400007	309 65 28
‡Sales Office is at 'F' Block, Unity Building, Narasimharaja Square, BANGALORE 560002	222 39 71

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AMENDMENT NO. 1 JANUARY 1981

TO

IS: 2026 (Part 1)-1977 SPECIFICATION FOR POWER TRANSFORMERS

PART 1 GENERAL

(First Revision)

Addenda

(*Page 15, clause 10.4.1*) - Add the following clause after 10.4.1:

'10.5 Cable Sealing Boxes - Cable sealing boxes for oil immersed transformers suitable for paper-insulated lead-sheathed cables for voltage class from 12 kV up to and including 36 kV shall comply with IS: 9147-1979***.'

(*Page 15, foot-note with '¶¶' mark*) - Add the following new foot-note after '¶¶' mark:

'*** Specification for cable sealing boxes for oil-immersed transformers suitable for paper-insulated lead-sheathed cables for highest system voltages from 12 kV up to and including 36 kV.'

(EDC 16)

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AMENDMENT NO. 2 JANUARY 1983

TO

IS : 2026 (Part I) - 1977 SPECIFICATION FOR POWER TRANSFORMERS

PART I GENERAL

(*First Revision*)

Alterations

(*Page* 29, *clause* **16.11.5**, *heading*) — Substitute the following for the existing heading:

'*Detection of Faults and the Evaluation of the Results of the Short-Circuit Test*'

(*Page* 29, *clause* **16.11.5.4**) — Substitute the following for the existing matter:

'**16.11.5.4** After completion of the tests, the transformer and the gas detection relay, if any, shall be inspected. The results of the short-circuit reactance measurements and the oscillograms taken during the different stages of the tests shall be examined for any indication of possible anomaly during the tests, especially any indications of change in the short-circuit reactance.'

Different procedures are followed at this stage for transformers of category 1 and category 2 or 3 as under:

a) *Transformers of category 1* — All the routine tests shall be repeated. The dielectric routine tests shall be at 75 percent of the original test value unless a higher value has been agreed between the manufacturer and the purchaser.

The transformer shall then be untanked for inspection of the core and windings, in order to reveal possible apparent defects, such as changes in lead position which, in spite of successful routine tests, might endanger the safe operation of the transformer.

The transformer is deemed to have passed the short-circuit tests if, firstly, the routine tests have been successfully repeated, secondly, the results of the short-circuit tests, measurements during short-circuit tests and out of tank inspection do not reveal any defects (displacements, deformations of windings, connections or supporting structures, or traces of discharge), and

thirdly, the short-circuit reactance measured after the tests differs from that measured in the original state by not more than:

two percent for transformers with circular concentric coils. However, for transformers having metal foil as a conductor in the low-voltage winding, higher values, not exceeding 4 percent for transformers with an impedance voltage of 3 percent or more, may be agreed between the manufacturer and the purchaser, or

7·5 percent for transformers with non-circular concentric coils having an impedance voltage of 3 percent or more. The value of 7·5 percent may be reduced by agreement between the manufacturer and the purchaser, but not below 4 percent.

NOTE — For transformers with non-circular concentric coils having an impedance voltage below 3 percent, the maximum variation in reactance cannot be specified in a general manner; practical knowledge of certain types of construction leads to the acceptance for such transformers of a variation equal to $(22{\cdot}5-5U_z)$ percent, U_z being the impedance voltage in percent.

If the three conditions for passing the short-circuit tests have been met, the transformer is restored to its original state and any further routine tests necessary to prove fitness for service are repeated before dispatch. If any of the three conditions have not been met, it may be necessary to dismantle the unit as far as is required to establish the cause of the change of the conditions.

b) *Transformers of category 2 and 3* — By agreement between the manufacturer and purchaser, a repeat of the routine tests, normally carried out at this point, may be postponed until after the inspection. The repeat of the dielectric routine tests shall be at 75 percent of the original test value unless a higher value has been agreed between the manufacturer and the purchaser.

NOTE — If the transformer was originally subjected to dielectric tests in accordance with Method 2 of IS : 2026 (Part III)-1981*, the voltage to be applied for the induced power-frequency overvoltage test should be subject to agreement between the manufacturer and the purchaser.

The transformer shall be untanked for inspection of the core and windings and is deemed to have passed the short-circuit tests if, firstly, the results of the short-circuit tests, measurements during short-circuit tests, measurement of short-circuit reactance and out of tank inspection do not reveal any apparent defects

*Specification for power transformers: Part III Insulation levels and dielectric tests (*second revision*).

(displacements, deformations of windings, connections or supporting structures, or traces of discharge) and, secondly, the routine tests have been successfully repeated. Agreement between the manufacturer and the purchaser is necessary concerning the interpretation of any differences in the reactance measurements.

If either of the two conditions for passing the tests has not been met, a more detailed examination may be required including, if necessary, a partial or complete dismantling of the unit.'

(*Page* 30, *clause* **16.11.6**) — Delete.

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AMENDMENT NO. 3 OCTOBER 1985

TO

IS:2026(Part 1)-1977 SPECIFICATION FOR POWER TRANSFORMERS

PART 1 GENERAL

(First Revision)

(*Page 9, Table 3, col 2, line 1*) - Substitute '4.5' *for* '4.0'.

(*Page 28, clause 16.11.4.4*) - Add the following new clause after this clause:

'16.11.4.5 In addition to the test of 16.11.4.4, when agreed between the purchaser and the supplier, a test may be carried out at specified value of short-circuit current in accordance with 8.1.1.2 and 9.1.2, for a duration of two seconds. This test shall be carried out after performing the test for dynamic ability to withstand short-circuit of 16.11.4.4. The asymmetry of the test current shall be as minimum as possible. The criteria for evaluation of test results shall be the same as that for the test to determine the dynamic ability to withstand short-circuit.'

(ETDC 16)

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